

NanoGT Match Report: Mucopolysaccharidosis type IV

Disease: Mucopolysaccharidosis type IV (ORPHA:578)

Primary gene: MCOLN1

Gene CDS: 1740 bp

Inheritance: Autosomal recessive

Target tissues: CNS, retina

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	ABO-101	AAV9	9.0/10	High	phase1/2
2	RGX-121	AAV9	9.0/10	High	phase3
3	CPCB-RPE1	AAV8	7.2/10	Medium	phase2/3
4	AVR-RD-01	LV	7.0/10	Medium	phase1/2
5	OAV101-IT	AAV9	7.0/10	Medium	approved

Match #1: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 9.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.5	1.0

Rationale

- Gene CDS 1740bp / cargo 4700bp (37% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal_storage
- Approval status: phase1/2

Match #2: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 9.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 1740bp / cargo 4700bp (37% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase3

Match #3: CPCB-RPE1

Precedent disease: Achromatopsia

Vector: AAV8

Tissue target: retina/photoreceptor

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	2.0	2.0
Protein class	1.5	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.7	1.0

Rationale

- Gene CDS 1740bp / cargo 4700bp (37% utilized)
- Vector tropism plus precedent target match: retina
- Both membrane proteins
- Inheritance match (Autosomal recessive <-> AR)

- Different pathway (lysosomal_storage vs retinal_phototransduction)
- Approval status: phase2/3

Match #4: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	0.3	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.5	1.0

Rationale

- Gene CDS 1740bp / cargo 4700bp (37% utilized)
- No tissue overlap (disease: ['CNS', 'retina'], vector: [])
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase1/2

Match #5: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	1.5	2.0
Tissue tropism	2.0	2.0
Protein class	1.0	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 1740bp / cargo 4700bp (37% utilized)

- Vector tropism plus precedent target match: cns
- Partial match: extracellular/secreted component present
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (lysosomal_storage vs motor_neuron)
- Approval status: approved